

Offchain Labs Glamsterdam Gas Parameters

Security Assessment (Summary Report)

October 19, 2026

Prepared for:
Harry Kalodner, Steven Goldfeder, and Ed Felten
Offchain Labs

Prepared by: **Simone Monica and Jaime Iglesias**

Table of Contents

Table of Contents 1
Project Summary 2
Project Targets 3
Executive Summary 4
About Trail of Bits 5
Notices and Remarks 6

Project Summary

Contact Information

The following project manager was associated with this project:

Mary O'Brien, Project Manager
mary.obrien@trailofbits.com

The following engineering director was associated with this project:

Benjamin Samuels, Engineering Director, Blockchain
benjamin.samuels@trailofbits.com

The following consultants were associated with this project:

Simone Monica, Consultant
simone.monica@trailofbits.com

Jaime Iglesias, Consultant
Jaime Iglesias@trailofbits.com

Project Timeline

The significant events and milestones of the project are listed below.

Date	Event
October 19, 2026	Delivery of final summary report

Project Targets

The engagement involved reviewing and testing the following target.

governance

Repository	https://github.com/ArbitrumFoundation/governance/
Version	`0b5d89306bb07f4d9f03ad645e67b840ff0f28e2`
Type	Solidity
Platform	EVM

Executive Summary

Engagement Overview

Offchain Labs engaged Trail of Bits to review the governance actions and proposal payload to update the ArbOS gas parameters on Arbitrum One and Nova in preparation for the Ethereum Glamsterdam upgrade.

A team of two consultants conducted the review from October 8 to October 9, 2026, for a total of four engineer-days of effort. With full access to source code and documentation, we performed static and dynamic testing of the codebase, using automated and manual processes.

Observations and Impact

The review covered the fork gate (which prevents changes from being implemented until Glamsterdam is live), the gas parameter update action, and the generated proposal payload. The review also focused on ensuring that updates cannot execute before Glamsterdam, that the selected values and precompile interfaces are correct, and that the payload calls the intended actions on the correct chains without unintended side effects.

No security issues were identified during the review.

About Trail of Bits

Founded in 2012 and headquartered in New York, Trail of Bits provides technical security assessment and advisory services to some of the world's most targeted organizations. We combine high-end security research with a real-world attacker mentality to reduce risk and fortify code. With 100+ employees around the globe, we've helped secure critical software elements that support billions of end users, including Kubernetes and the Linux kernel.

We maintain an exhaustive list of publications at https://github.com/trailofbits/publications, with links to papers, presentations, public audit reports, and podcast appearances.

In recent years, Trail of Bits consultants have showcased cutting-edge research through presentations at CanSecWest, HCSS, Devcon, Empire Hacking, GrrCon, LangSec, NorthSec, the O'Reilly Security Conference, PyCon, REcon, Security BSides, and SummerCon.

We specialize in software testing and code review assessments, supporting client organizations in the technology, defense, blockchain, and finance industries, as well as government entities. Notable clients include HashiCorp, Google, Microsoft, Western Digital, Uniswap, Solana, Ethereum Foundation, Linux Foundation, and Zoom.

To keep up with our latest news and announcements, please follow @trailofbits on X or LinkedIn and explore our public repositories at https://github.com/trailofbits. To engage us directly, visit our "Contact" page at https://www.trailofbits.com/contact or email us at info@trailofbits.com.

Trail of Bits, Inc.
228 Park Ave S #80688
New York, NY 10003
https://www.trailofbits.com
info@trailofbits.com

Notices and Remarks

Copyright and Distribution

© 2026 by Trail of Bits, Inc.

All rights reserved. Trail of Bits hereby asserts its right to be identified as the creator of this report in the United Kingdom.

Trail of Bits considers this report public information; it is licensed to Offchain Labs under the terms of the project statement of work and has been made public at Offchain Labs' request. Material within this report may not be reproduced or distributed in part or in whole without Trail of Bits' express written permission.

The sole canonical source for Trail of Bits publications is the Trail of Bits Publications page. Reports accessed through sources other than that page may have been modified and should not be considered authentic.

Test Coverage Disclaimer

Trail of Bits performed all activities associated with this project in accordance with a statement of work and an agreed-upon project plan.

Security assessment projects are time-boxed and often rely on information provided by a client, its affiliates, or its partners. As a result, the findings documented in this report should not be considered a comprehensive list of security issues, flaws, or defects in the target system or codebase.

Trail of Bits uses automated testing techniques to rapidly test software controls and security properties. These techniques augment our manual security review work, but each has its limitations. For example, a tool may not generate a random edge case that violates a property or may not fully complete its analysis during the allotted time. A project's time and resource constraints also limit their use.